

THE EMERGENCE AND ROLE OF ARTIFICIAL INTELLIGENCE IN MODERN LIBRARIANSHIP EDUCATION IN NIGERIA: A SCOPING REVIEW

Abstract

Artificial Intelligence (AI) is no longer something that belongs only in science fiction or advanced computer laboratories. Today, AI tools are becoming part of everyday life, including in education, business, healthcare, and information management. In the field of Library and Information Science (LIS), AI is changing how libraries operate and how librarians are trained.

This scoping review examines how AI is emerging in librarianship education in Nigeria, how it is currently being used or discussed, and what role it plays in preparing future librarians. The study draws from existing research on AI awareness, AI literacy, curriculum development, and technological readiness in Nigerian library schools.

Findings show that although awareness of AI is growing among LIS educators and librarians, structured integration into curricula remains limited. Major challenges include inadequate infrastructure, limited training opportunities, and absence of strong institutional policies.

However, there are significant opportunities for curriculum reform, educator capacity building, and strategic partnerships. This review concludes that AI must become a central part of modern librarianship education in Nigeria if the profession is to remain relevant in a digital and data-driven world.

1. INTRODUCTION

Over the past two decades, higher education around the world has experienced rapid digital transformation. Technologies such as cloud computing, big data, automation tools, and artificial intelligence are now shaping how students learn and how professionals are trained. Artificial Intelligence, in particular, has gained attention because it allows computers and systems to perform tasks that normally require human intelligence, such as learning from data, recognizing patterns, making predictions, and even generating text and images.

In the context of libraries, AI can assist in cataloguing materials automatically, answering users' questions through chatbots, recommending books based on reading history, and analyzing user behavior to improve services. However, beyond library operations, an important question arises: how are future librarians being prepared to understand, use, and manage AI tools?

In Nigeria, Library and Information Science education plays a crucial role in training professionals who will manage academic, public, school, and special libraries. As the information environment becomes more digital and automated, LIS education must evolve to reflect these changes.

Scholars have begun to examine AI awareness and readiness among Nigerian librarians and educators (Adeleke & Olorunsola, 2023; Yusuf & Abdullahi, 2024). Yet, there is still limited comprehensive discussion focusing specifically on AI in librarianship education.

This scoping review therefore explores the emergence, current status, challenges, and future role of Artificial Intelligence in modern librarianship education in Nigeria.

2. Understanding Artificial Intelligence in Simple Terms.

Artificial Intelligence refers to computer systems that can perform tasks that normally require human thinking. These tasks include understanding language, solving problems, making decisions, and learning from experience. AI systems include:

- Machine learning systems that learn from data
- Natural language processing tools that understand human language.
- Chatbots that answer questions automatically.
- Recommendation systems that suggest books or materials.
- Predictive tools that analyze trends in education.

AI can support teaching by grading assignments, creating personalized learning materials, answering students' questions, and helping with research tasks. In libraries, AI can organize information faster and more accurately than traditional manual systems.

Therefore, for librarianship education to remain modern and relevant, students must understand both how AI works and how to use it responsibly.

3. AI Awareness in Nigerian Librarianship Education

Research: shows that awareness of AI among Nigerian librarians is increasing. For example, studies examining university librarians reveal that many professionals understand that AI can improve efficiency and service delivery (Adeleke & Olorunsola, 2023).

Librarians generally believe that AI can help reduce repetitive tasks and allow professionals to focus on more intellectual responsibilities.

However, awareness does not automatically mean integration. While librarians may know about AI tools like chatbots and automated cataloguing systems, many LIS programs in Nigeria have not fully incorporated AI courses into their curriculum (Yusuf & Abdullahi, 2024).

In many library schools, traditional courses such as cataloguing, classification, indexing, reference services, and collection development remain dominant. Although these subjects are still important, they must now be taught alongside digital skills, data literacy, and AI understanding.

4. Curriculum Integration: Current Reality

One major issue in Nigerian LIS education is the slow pace of curriculum reform. While some universities have introduced courses related to Information Technology, digital libraries, and database management, specific and detailed AI-focused courses remain rare.

According to research on LIS educators' perceptions, many lecturers support the idea of integrating AI into library school curricula but feel they lack sufficient training to teach such subjects confidently (Okoye & Nwosu, 2024). This creates a gap between intention and implementation.

In addition, curriculum updates in Nigerian universities often require regulatory approval and administrative processes, which can delay the introduction of emerging subjects. As a result, many LIS students graduate with theoretical knowledge of technology but limited hands-on experience with AI tools.

5. Role of AI in Teaching and Learning

Artificial Intelligence can play multiple roles in librarianship education:

5.1 Supporting Teaching

AI tools such as language models, research assistants, and digital tutors can help lecturers prepare teaching materials, generate examples, and provide quick explanations. AI can also assist in grading and providing feedback to students.

5.2 Supporting Students

Students can use AI tools to:

- Search for research materials more effectively
- Generate summaries of academic articles
- Improve academic writing
- Practice coding or data analysis

However, ethical guidance is necessary to ensure students use AI responsibly and avoid academic misconduct.

5.3 Practical Training

Practical training helps LIS students apply AI knowledge in real library settings. Students can use AI-powered tools for automated cataloguing, metadata generation, and intelligent search systems, gaining hands-on experience with tasks they will encounter in modern libraries.

AI systems used in modern libraries such as automated cataloguing systems, and smart discovery platforms can be introduced in laboratory sessions. This experience builds confidence, problem-solving skills, and digital competence, bridging the gap between theory and practice (Yusuf & Abdullahi, 2024).

6. Challenges Facing AI Integration in Nigerian LIS Education

Despite the opportunities, several challenges remain, such as:

6.1 Infrastructure Limitations

Reliable internet access, updated computer laboratories, and licensed software are not always available in many Nigerian universities. Poor infrastructure makes practical AI training difficult (Adeleke & Olorunsola, 2023).

6.2 Limited Training for Educators

Many LIS educators were trained before AI became widespread. Without professional development workshops and training programs, it is difficult for them to teach AI confidently (Okoye & Nwosu, 2024).

6.3 Funding Constraints

AI systems and digital infrastructure require investment. Limited funding in higher education affects the ability of library schools to adopt advanced technologies.

6.4 Ethical and Policy Concerns

There is still limited national policy direction guiding AI use in Nigerian higher education. Clear frameworks are needed to address data privacy, algorithmic bias, and responsible use.

7. Opportunities and Future Directions

Despite these challenges, the future of AI in librarianship education in Nigeria is promising. AI offers several opportunities for Nigerian LIS education. Universities can introduce AI courses, train lecturers through workshops, and build partnerships with technology companies. Encouraging student research on AI can also strengthen skills and employability (Adeleke & Olorunsola, 2023; Okoye & Nwosu, 2024; Yusuf & Abdullahi, 2024).

7.1 Curriculum Reform

Universities can gradually introduce modules such as:

- Introduction to Artificial Intelligence
- AI Applications in Libraries
- Data Analytics for Information Professionals
- Ethical Issues in AI

7.2 Capacity Building

Regular workshops, conferences, and online training can improve educators' confidence and competence in AI topics. Improving AI skills among LIS educators and students is essential. Regular workshops, seminars, and online training can help lecturers gain confidence in teaching AI and allow students to practice using AI tools effectively (Okoye & Nwosu, 2024).

7.3 Partnerships

Collaborating with technology companies, international LIS organizations, and research institutions can give Nigerian LIS students and educators access to AI tools, practical training, and new learning opportunities (Yusuf & Abdullahi, 2024).

7.4 Research Development

Encouraging postgraduate research in AI and librarianship can strengthen local scholarship and provide Nigerian perspectives in global discussions.

Encouraging students and faculty to conduct research on AI applications in libraries can strengthen local expertise, contribute to innovation, and provide solutions tailored to Nigerian library environments (Adeleke & Olorunsola, 2023).

8. CONCLUSION

Artificial Intelligence is transforming how information is created, stored, accessed, and used. For librarianship education in Nigeria to remain relevant, it must adapt to this transformation. Although awareness of AI is growing, structured curriculum integration, educator training, and infrastructure investment are still needed.

The emergence of AI in librarianship education is not just about teaching students to use new tools. It is about preparing future librarians to think critically about technology, manage digital systems responsibly, and lead innovation in Nigerian libraries.

If Nigerian LIS programs embrace AI thoughtfully and strategically, they can produce graduates who are confident, skilled, and ready for the modern information environment.

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